Foggy Bottom Mystery: GDD V2

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Introduction

Game Summary Pitch

Have you ever been to Colgate University's Foggy Bottom Observatory?

Me neither.

What is this place, really? What is going on in there?? Why is it called Foggy Bottom???

A newly hired Astronomy Professor arrives at Colgate, eager and naive. He hears whispers about the observatory - an old dome tucked away on the edge of campus, built in the 1950s. Students avoid it. Faculty don't talk about it. One night, curiosity wins. He walks in. The floor gives. He falls.

He wakes up at the bottom - legs broken, surrounded by fog and old brick. The only way out is up. Luckily, there's a hook and a rope.

To the foggy bottom...

Inspiration

Professor Layton

Professor Layton acts as an inspiration for the visual and musical elements. It has a Japanese-European 2D animation style and a whimsical & classical soundtrack that captures the player's attention. The cadence of the music and the play is also a huge part of the game.

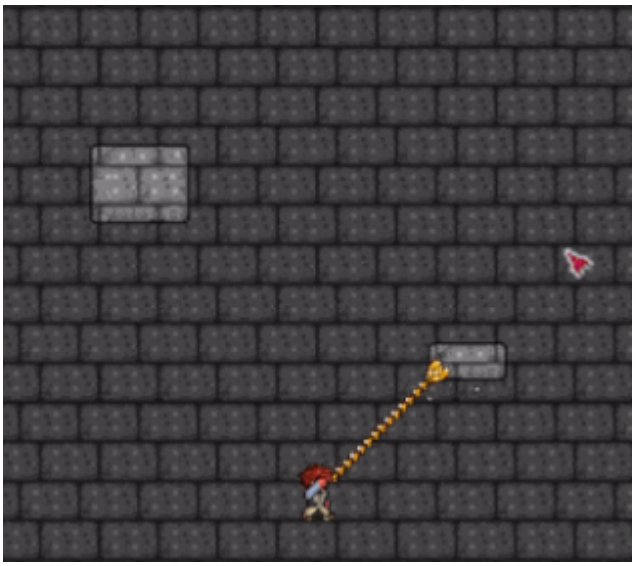
Getting Over It with Bennett Foddy

Getting Over It (GOI) is famous for its unintuitive and frustrating mechanics. It is also rather infamous for its level design that doesn't guarantee consistent progress — you can climb to right before the finish line

and still fall all the way back to the very beginning with just one misclick. The dry, passive-aggressive voice-over when the player falls off a platform also adds to the fun of the game.

Terraria

Terraria has a hooking mechanism that lets the user move between platforms using a hook item. The idea of using such a mechanism in a 2D game is very appealing to me, and I think it is much more interesting than just jumping.



Player Experience

The player wakes up at the bottom of a dark, foggy shaft. No jumping. No walking. Just a hook and a rope.

Left click to shoot. Swing with the mouse. Reel in to climb. Miss, and you fall - sometimes all the way back down. The hook is your only friend, and the observatory doesn't want you to leave.

Reflective surfaces bounce your hook. Crumbling walls won't hold. Every throw is a gamble. It's frustrating, funny, and deeply satisfying when it clicks.

Platform

The game is developed to be released on Mac & Windows PC with the Pyray framework.

Software

- VS Code
- DaVinci Resolve (audio editing, image editing)
- Excali Draw (color pipette / picker)

Genre

Singleplayer, platformer, climbing

Target Audience

Colgate University COSC 481 students

Concept

Gameplay Overview

The player controls an Astronomy Professor, legs broken from the fall, who must climb out of the Foggy Bottom Observatory shaft using nothing but a hook and a rope. No walking. No jumping. Just throwing, swinging, and praying the brick holds.

Theme Interpretation

Foggy background, Climb / Vertical Levels: a whimsical interpretation of the origin of the name "Foggy Bottom Observatory"

Inability to walk or jump: the character is injured from the fall.

Primary Mechanics

Mechanic	Description
Hook	Left click shoots a hook toward the mouse cursor. It sticks to anchor points, bounces off walls. Left click again to release.
Reel in/out	W / Up reels the rope in, S / Down lets it out
Swing	A / D pumps the swing left and right while hooked. Pendulum physics with constraint projection.

Secondary Mechanics

Mechanic	Description
Momentum	Releasing the hook preserves the player's velocity - build swing speed to launch to the next anchor
Wall collision	Player slides along walls while swinging or falling, cannot walk
Reflect	There are surfaces where the hook gets reflected rather than getting hooked (L2)

Art

Theme Interpretation

The player is trapped inside a bricked shaft with thick fog rolling behind them. The character should look adventurous yet academic - a professor who wasn't dressed for this.

The character sprite is an archaeologist (although he is an astronomy professor) with whip animations that naturally map to the hook/rope mechanic. Animations: idle, movement, roll, whip attack, whip AoE, shoot, damage, death.

Design

Old, weathered brick - the observatory was built in the 1950s and it shows. 2D Japanese-English animation style to heighten the mysterious, puzzling nature of the situation.

Asset Candidates:

Asset	Source	License
2D Pixel Art Archaeologist Sprites (character)	Elthen - itch.io	Free (commercial/non-commercial)
Brick 2D Tileset 16x16 (walls/platforms)	DeadlyEssence01 - OpenGameArt	CC0
2D Dungeon Wall 32px Brick Brown	lukems-br - OpenGameArt	CC0

Audio

Music

Whimsical & cadenced, like Professor Layton. The music should feel like you're solving something - not like you're in danger. Light tension, not dread.

Music Candidates:

Track	Source
"페리온 OST"	Maple Story

Sound Effects

Dramatic sound effect when the player falls more than a certain y distance. "Hooking" sound when the hook latches (nail-into-wall). Metallic clank when reflected off a surface.

SFX Candidates:

Sound	Source	License
Grappling Hook (launch)	16bitstudios - Freesound	CC0

Sound	Source	License
Metal Clank (hook impact / reflect)	JustInvoke - Freesound	CC-BY 4.0
Rope Under Tension (swing)	gear_clinkz - Freesound	CC0
Whoosh (falling)	Kinoton - Freesound	CC0
Body Fall / Thud (dramatic fall)	Breviceps - Freesound	CC0

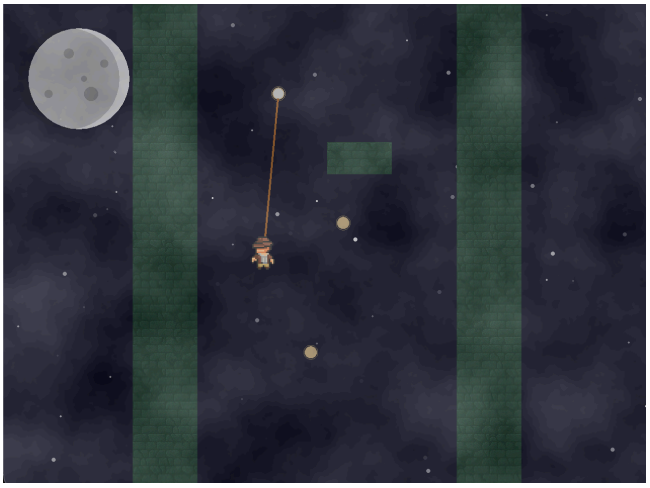
Game Experience

The professor wakes up at the bottom of the observatory shaft. Legs broken. Fog everywhere. He can't walk. He can't jump. But there's a hook and a rope lying next to him - and a long, long way up.

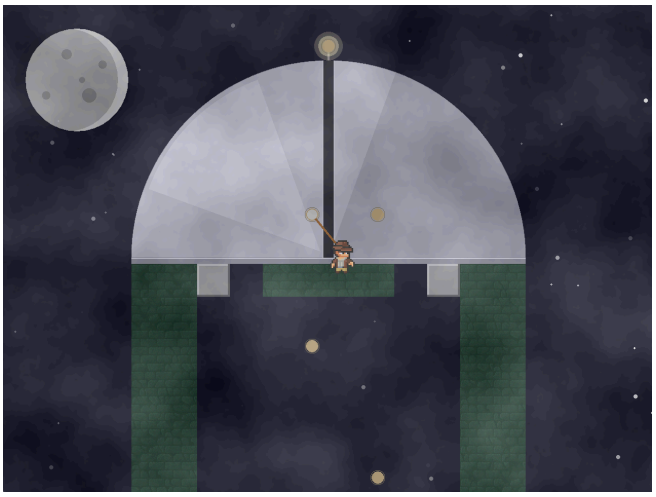
In-game Screenshots

Start

Level 1



Level 2



End

UI

I want the UI to feel similar to that of `Professor Layton` — a 2010s 2D animation style with the nuance of early-1900s attire.

Controls

Mouse

Left Click: shoot hook toward cursor / release hook when attached

Mouse Position: aim direction for hook

Keyboard

W / Up: reel in rope

S / Down: let out rope

A / D: pump swing left / right

P: pause (shows control reference)

E: Fly mode & debug overlay (FPS, position, velocity, hook state)

Technical Details

Swing Mechanics

The hook uses **position-based constraint projection** instead of angular pendulum math. Each frame:

1. The player moves freely under gravity + A/D swing input.
2. If the new position drifts farther than `rope_length` from the anchor, snap it back onto the rope's "circle" — the imaginary circle of all points exactly `rope_length` from the anchor.
3. Decompose velocity into a radial component (along the rope) and a tangential component (perpendicular). Kill only the *outward* radial part — what's left is the swing.

The result feels like a real rope: you can pull yourself in, let yourself out, and your sideways momentum carries you around the anchor. No trigonometry, just a vector projection per frame.

Reference: Swinging Physics for Player Movement.

Perlin Noise as Fog & Alpha Masking

The fog overlay is generated at startup, not loaded from a file. Two raylib helpers do the work:

- `gen_image_perlin_noise(...)` produces a tileable greyscale cloud pattern.
- `image_alpha_mask(image, mask)` writes the mask's brightness into the image's alpha channel.

So a flat grey image gets the noise pattern carved into its alpha — bright noise = visible fog, dark noise = clear. The texture is then drawn with `TEXTURE_WRAP_MIRROR_REPEAT` and a scrolling source rect, so the fog drifts horizontally without a visible seam.

The same trick recolors the brick sprite: load the brown bricks, convert to greyscale, then use them as an alpha mask over a green canvas. The brick *pattern* survives (via alpha) while the *colour* becomes the observatory's green.

References: Image generation, Alpha masking.

Development Timeline

Minimum Lovable Product

#	Assignment	Type	Status	Finish By	Notes
1	Initial Design Doc	doc	🟢	April 16, 10 PM	
2	Code Commit V1	dev	🟢	April 19, 10 PM	L1 playable with hook mechanics
3	Final Code	dev	🟢	April 30, 5 PM	L2, ending sequence
4	Final Design Doc	doc	🟢	May 4, 10 PM	
5	Demo Vid	doc	🟢	May 4, 10 PM	
6	Presentation	other	🔴	May 6, ??:??	

References

- COSC481 Playful Thinking, Serious Coding at Colgate University
- Thanks to Professor Elodie Fourquet (efourquet@colgate.edu) for guidance and invaluable feedback.
- Thanks to Edgar (esuritis@colgate.edu) for showing his dev mode (god mode).

- Owl Chemist GDD
- Foggy Bottom Observatory
- `.github` template
- Perlin Noise
- Alpha Masking
- Swing Mechanics

*AI was used for grammtical check & alpha masking implementation